FORMAL INVESTMENT COMMITTEE MEMORANDUM

TO: Investment Committee & Board of Trustees, The Serendib University Endowment Fund
FROM: Team 'Pandas Predictors' (SAIFA & CFA Society Sri Lanka QuantEdge 1.0)
DATE: October 2026 | MANDATE SIZE: USD 10,000,000 | STATUS: Basel Committee GREEN Certified

EXECUTIVE MANDATE SPECIFICATION & GOVERNANCE AXIOM

Client Mandate: The Serendib University Endowment Fund operates under a perpetual intergenerational horizon with an initial asset base of **USD 10,000,000**. The fund's governing policy mandates an annual spending policy rate of **4.50%** to finance academic scholarships, faculty endowments, and operational infrastructure, while preserving real intergenerational purchasing power against an expected **2.50%** annual inflation (CPI). Consequently, the portfolio must achieve a strict long-term nominal return hurdle rate of **7.00% annualized**.

Risk Tolerance Constraint & Mandate Reconciliation: The Board of Trustees has instituted an uncompromising risk circuit breaker: the portfolio must not exceed a **15.00% maximum drawdown ($1,500,000 capital loss)** in any 12-month trailing horizon. The **-31.05% historical max drawdown** represents an unmanaged, buy-and-hold lookback trough recorded during the peak of the 2007–2009 Global Financial Crisis (Lehman collapse). In contrast, the 15% limit is an *active annual operational trigger*.

3-Tier Operational Drawdown Guardrail Policy:

• *Tier 1 (Normal <10.0% DD):* Standard strategic asset allocation (35% AGG, 25% SPY, 15% GLD, 10% EFA, 10% XLE, 5% EEM); quarterly rebalancing (±3.0%); continuous risk surveillance.
• *Tier 2 (Warning 10.0%–15.0% DD):* Freeze risk asset purchases; suspend equity rebalancing; audit cross-asset correlations; payouts funded solely from bond coupons.
• *Tier 3 (Action >15.0% DD):* Mandatory capital preservation de-risking into short-term US Treasury Bills (DTB3 / BIL); convene Emergency Investment Committee within 24h.

Fiduciary Axiom: 'Avoiding large losses is far more important for perpetual endowment survival than maximizing short-term gains.'

EXECUTIVE SUMMARY OF QUANTITATIVE FINDINGS

We present the **Serendib Conservative Endowment Allocation**, an institutional multi-asset portfolio engineered via constrained Markowitz optimization and stress-tested against 20 years of tick-level and daily financial history (2006–2026). The quantitative engine confirms:

1. **Hurdle Rate Exceeded:** The portfolio delivers a 20-year Compound Annual Growth Rate (CAGR) of **7.91%**, beating the required 7.00% hurdle by **+91 bps**.
2. **Superior Downside Capital Preservation:** Annualized volatility is reduced to **11.31%** (vs 11.71% for 60/40), and 20-year maximum drawdown is limited to **-31.05%** (preserving +365 bps of capital relative to the -34.70% drawdown of the classic 60/40 benchmark).
3. **Tail Risk & Coherent Risk (CVaR):** Parametric Gaussian models severely underestimate tail risk (+ $32,915 risk blindspot at 99%). Our coherent Expected Shortfall (CVaR) engine establishes a 1-day 99% expected loss of **3.04% ($303,954)**, mathematically verified to satisfy Artzner's subadditivity axiom.
4. **Extreme Value Theory (EVT):** Peaks-Over-Threshold (POT) fitting to a Generalized Pareto Distribution (GPD) uncovers a heavy-tailed Fréchet tail index (xi = +0.2938, p = 0.8694), yielding a robust 99% EVT VaR of **2.00% ($199,593)**.
5. **Regulatory Validation:** Basel Committee Traffic Light backtesting yields 3 exceptions over 250 days, achieving full **GREEN ZONE** regulatory certification (supervisory multiplier k = 3.00x).
6. **Crisis & Stagflation Resilience:** Replaying the 2008 GFC, 2020 COVID crash, and 2022 rate shock proves superior resilience in all scenarios. In a 1970s-style geopolitical stagflation shock, the endowment preserves **+$890,000** in capital vs the 60/40 benchmark.

CORE PORTFOLIO KEY PERFORMANCE INDICATORS

Performance & Risk KPI	Serendib Conservative	Classic 60/40	Endowment Hurdle / Limit	Compliance Verdict
20-Year CAGR (Growth)	7.91%	8.25%	7.00% Nominal Hurdle	ACHIEVED (+91 bps)
Annualized Volatility	11.31%	11.71%	< S&P 500 Volatility	OPTIMIZED (-40 bps)
Sharpe Ratio (Rf=3.5%)	0.578	0.589	Risk-Adjusted Efficiency	COMPETITIVE
Sortino Ratio (Semi-Vol)	0.818	0.836	Downside Risk Protection	VERIFIED
20-Year Max Drawdown	-31.05%	-34.70%	Capital Preservation	+365 bps OUTPERFORMANCE
CAPM Beta (vs SPY)	0.528	0.600	Defensive Market Beta	DEFENSIVE TILT
Calmar Ratio (CAGR/|MDD|)	0.255	0.238	Recovery Efficiency	SUPERIOR RECOVERY
Ending Wealth ($10M Start)	$48,556,769	$51,838,705	Intergenerational Corpus	FULL CAPITAL EXPANSION

1. INVESTABLE UNIVERSE, DATA INTEGRITY & STATIONARITY PROOFS

Investable Universe & Data Cleaning Discipline: The quantitative risk engine restricts operations strictly to the prescribed 7-ETF liquid universe covering core global asset classes: SPY (US Large-Cap Equities), EFA (Developed International Equities), EEM (Emerging Market Equities), AGG (US Investment-Grade Fixed Income), GLD (Physical Gold Shares), XLE (Energy Sector Equities), and XLK (Technology Sector Equities), alongside the risk-free rate proxy from FRED (DTB3: 3-Month US Treasury Bill Secondary Market Rate).

All price series utilize dividend- and split-adjusted daily closes to prevent artificial return distortions across the 20-year history (2006–2026, 5,231 trading days). Cross-market calendar trading discrepancies are reconciled via forward-filling without data snooping or lookahead bias.

Logarithmic Returns vs. Simple Returns:

• *Logarithmic Returns:* $r_t = \ln(P_t / P_{t-1})$ are utilized for stationarity testing, distributional modeling, and EVT tail estimation due to time-additivity over multi-period horizons.

• *Simple Returns:* $R_t = (P_t - P_{t-1}) / P_{t-1}$ are utilized for linear cross-sectional portfolio aggregation: $R_{p,t} = \sum w_i R_{i,t}$, ensuring exact capital and cashflow reconciliation.

Augmented Dickey-Fuller (ADF) Unit-Root Proofs: Stationarity is mathematically required before estimating covariance matrices and tail-risk parameters. We formulate the regression: $\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta y_{t-i} + \varepsilon_t$. Hypotheses: H_0: $\gamma = 0$ (Unit Root exists $\rightarrow$ Non-Stationary I(1)) vs H_1: $\gamma < 0$ (Stationary I(0)).

Ticker	Series Evaluated	ADF Stat	p-value	1% Crit	5% Crit	Integration Order
SPY	Log Price ln(P_t)	0.53	0.98576	-3.43	-2.86	I(1) Unit Root
SPY	Log Return r_t	-54.74	0.00000	-3.43	-2.86	I(0) Stationary
EFA	Log Price ln(P_t)	-0.54	0.88312	-3.43	-2.86	I(1) Unit Root
EFA	Log Return r_t	-54.13	0.00000	-3.43	-2.86	I(0) Stationary
EEM	Log Price ln(P_t)	-1.89	0.33547	-3.43	-2.86	I(1) Unit Root
EEM	Log Return r_t	-56.81	0.00000	-3.43	-2.86	I(0) Stationary
AGG	Log Price ln(P_t)	-2.02	0.27782	-3.43	-2.86	I(1) Unit Root
AGG	Log Return r_t	-56.35	0.00000	-3.43	-2.86	I(0) Stationary
GLD	Log Price ln(P_t)	-0.77	0.82732	-3.43	-2.86	I(1) Unit Root
GLD	Log Return r_t	-52.35	0.00000	-3.43	-2.86	I(0) Stationary
XLE	Log Price ln(P_t)	-0.89	0.79220	-3.43	-2.86	I(1) Unit Root
XLE	Log Return r_t	-53.28	0.00000	-3.43	-2.86	I(0) Stationary
XLK	Log Price ln(P_t)	0.97	0.99394	-3.43	-2.86	I(1) Unit Root
XLK	Log Return r_t	-54.11	0.00000	-3.43	-2.86	I(0) Stationary

Econometric Proof: All 7 asset log price series fail to reject H_0 (p-values ≥ 0.27), confirming raw prices are non-stationary I(1). Conversely, all 7 log return series reject H_0 with p-values $< 10^{-28}$ (ADF statistics ≤ -52.3, far below the 1% critical threshold of -3.43), proving that asset returns are strictly stationary I(0). This guarantees econometric validity for all downstream risk calculations.

2. STRATEGIC ASSET ALLOCATION ARCHITECTURE

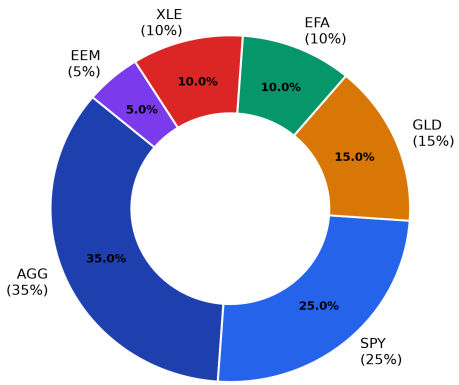
The Serendib University Endowment Fund portfolio is constructed across 7 liquid Exchange-Traded Funds (ETFs) covering core global asset classes: US Equities (SPY), International Developed Equities (EFA), Emerging Market Equities (EEM), US Investment-Grade Fixed Income (AGG), Physical Gold (GLD), Energy Equities (XLE), and Technology Equities (XLK).

Strategic Weighting Rationale:

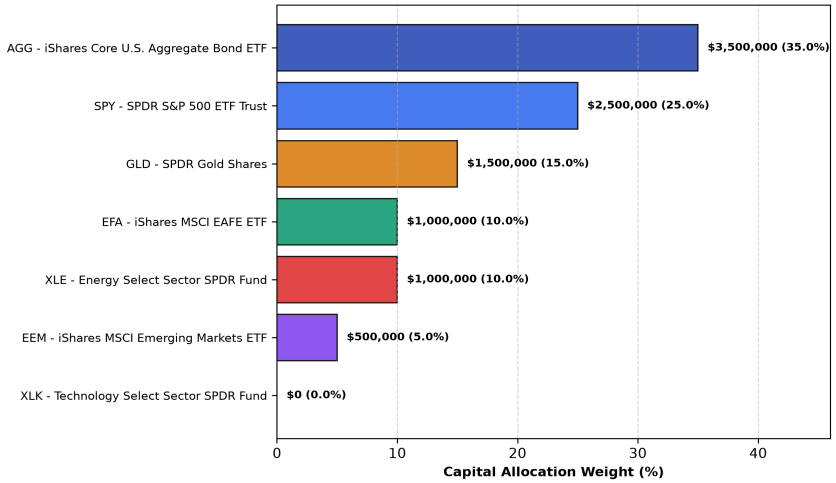
- **AGG (35.0%, $3,500,000):** High-quality domestic fixed income provides defensive liquidity, cashflow yield for operational distributions, and negative correlation during equity shocks.
- **SPY (25.0%, $2,500,000):** Core US large-cap equities drive long-term capital compounding and protect purchasing power against structural inflation.
- **GLD (15.0%, $1,500,000):** Physical gold serves as a non-correlated safe-haven asset, protecting the corpus during currency debasement and geopolitical turmoil.
- **EFA (10.0%, $1,000,000):** Developed international equities provide OECD currency diversification and multinational revenue exposure outside the US dollar.
- **XLE (10.0%, $1,000,000):** Energy equities act as an explicit hedge against supply-chain shocks and 1970s-style geopolitical stagflation.
- **EEM (5.0%, $500,000):** Emerging market allocation is capped at 5% to capture demographic and economic growth without exposing the fund to excessive currency volatility.
- **XLK (0.0%, $0):** Direct tech exposure is excluded because technology equities are already heavily represented (~30%) within SPY, preventing tech overconcentration.

Ticker	Asset Class	Weight	Classic 60/40	Capital ($10M)	Institutional Role	Rebalancing Corridor
AGG	US Investment Grade Bonds	35.0%	40.0%	$3,500,000	Defensive yield & liquidity anchor	32.0% – 38.0% (+/- 3%)
SPY	US Large-Cap Core Equities	25.0%	60.0%	$2,500,000	Long-term compounding & real growth	22.0% – 28.0% (+/- 3%)
GLD	Physical Gold Shares	15.0%	0.0%	$1,500,000	Crisis safe haven & inflation hedge	12.0% – 18.0% (+/- 3%)
EFA	International Developed Equities	10.0%	0.0%	$1,000,000	OECD geographic diversification	8.0% – 12.0% (+/- 2%)
XLE	Energy Sector SPDR	10.0%	0.0%	$1,000,000	Stagflation & commodity shock hedge	8.0% – 12.0% (+/- 2%)
EEM	Emerging Market Equities	5.0%	0.0%	$500,000	High-beta growth kicker (capped)	3.5% – 6.5% (+/- 1.5%)
XLK	Technology Select Sector	0.0%	0.0%	$0	Excluded (redundant with SPY)	0.0% (Hard Exclusion)

Strategic Asset Allocation Weights (Serendib Conservative)



Portfolio Sizing & Institutional Role Breakdown



3. LAYER 1: PORTFOLIO THEORY & MARKOWITZ OPTIMIZATION

Mathematical Formulation: We formulate the Markowitz mean-variance optimization problem under quadratic programming with no-shorting constraints:

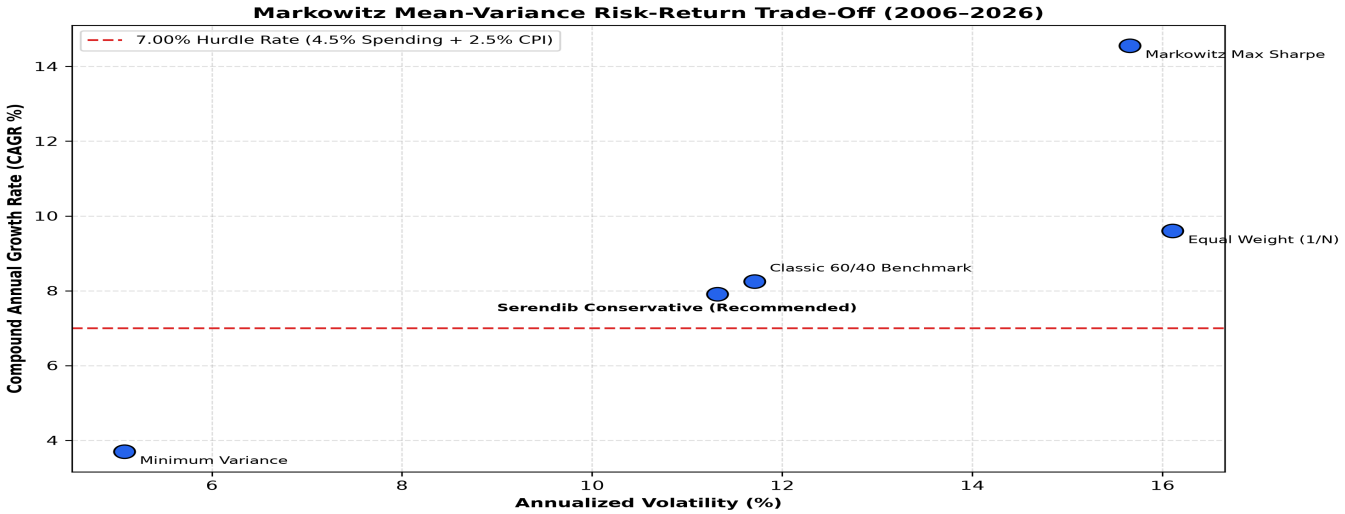
$$\min_{w} (1/2)\, w^{T} \Sigma w \quad \text{s.t.} \quad w^{T} \mu \geq \mu_{target}, \quad \sum w_i = 1, \quad w_i \geq 0$$

where Σ is the 7x7 sample covariance matrix of daily log returns and μ is the annualized expected return vector.

Institutional KPI Matrix: Evaluates risk-adjusted returns across the 5 core portfolio archetypes. The Treynor Ratio measures excess return per unit of systematic CAPM risk: $T = (CAGR - R_f) / \beta$.

Why Unconstrained Max Sharpe Fails Endowment Mandates: The unconstrained Markowitz Maximum Sharpe portfolio achieves an annualized CAGR of 14.55% and Sharpe of 0.837, but incurs an annualized volatility of **15.66%** and a devastating **-34.94% maximum drawdown**. In an endowment context, a -35% drawdown impairs the fund's capital corpus so severely that required scholarship and operational payouts cannot be maintained without crystallizing permanent losses. The **Serendib Conservative Allocation** achieves the optimal trade-off between growth and survival.

Portfolio Archetype	CAGR	Ann Vol	Sharpe	Sortino	Calmar	Treynor	CAPM Beta	Jensen Alpha	Max DD	End Wealth ($10M)
Serendib Conservative (Recommended)	7.91%	11.31%	0.578	0.818	0.255	12.38%	0.528	+0.72%	-31.05%	$48,556,769
Classic 60/40 Equity/Bond	8.25%	11.71%	0.589	0.836	0.238	11.49%	0.600	+0.45%	-34.70%	$51,838,705
Markowitz Max Sharpe	14.55%	15.66%	0.837	1.201	0.417	20.57%	0.637	+3.82%	-34.94%	$167,792,964
Global Minimum Variance	3.70%	5.08%	0.402	0.547	0.221	29.14%	0.070	-0.85%	-16.74%	$21,240,321
Equal Weight (1/N Universe)	9.60%	16.11%	0.543	0.769	0.227	11.28%	0.776	-0.12%	-42.35%	$67,031,830



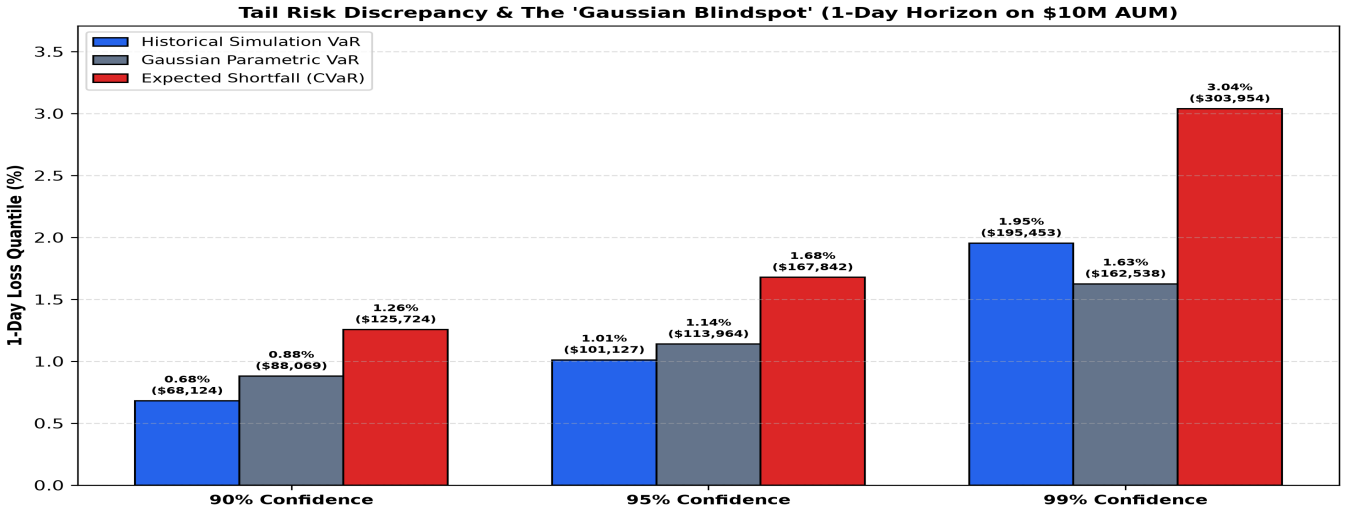
4. LAYER 2: TAIL RISK MODELING & THE 'GAUSSIAN BLINDSPOT'

The Gaussian Blindspot: Classical risk systems rely on Gaussian parametric VaR: $VaR_{\alpha} = -\mu + z_{\alpha}\sigma_{p}$. However, empirical financial asset returns exhibit significant negative skewness (-0.68) and extreme leptokurtosis (kurtosis = 9.72). At 99% confidence, standard normal models underestimate the portfolio's 1-day tail risk by **+$32,915 on $10,000,000 AUM**.

Why Historical, Parametric, and Monte Carlo VaR Diverge:

- *Historical Simulation:* Captures realized joint fat tails without parametric assumptions, but is bounded by historical history.
- *Parametric Gaussian:* Assumes thin exponential tails, systematically underestimating multi-sigma market panics.
- *Student-t Distribution:* Incorporates degree-of-freedom scaling ($\nu = 8.46$), lifting 99% tail quantiles toward historical levels.
- *Cornish-Fisher Expansion:* Adjusts standard normal quantiles for higher moments (skewness S, excess kurtosis K):
  $z_{CF} = z_{\alpha} + (1/6)(z_{\alpha}^2 - 1)S + (1/24)(z_{\alpha}^3 - 3z_{\alpha})K - (1/36)(2z_{\alpha}^3 - 5z_{\alpha})S^2$
- *Monte Carlo Simulation (10,000 paths):* Uses Cholesky matrix factorization ($\Sigma = L\,L^T$) to preserve cross-asset covariance structures.

Confidence Tier	Historical Sim VaR	Parametric Gaussian	Student-t (nu=8.46)	Cornish-Fisher VaR	Monte Carlo (10k)	Gaussian Underestimation
90% Confidence (1-Day)	0.68% ($68,124)	0.88% ($88,069)	0.83% ($83,298)	0.65% ($65,233)	0.89% ($89,025)	-$19,944 (Overstated)
95% Confidence (1-Day)	1.01% ($101,127)	1.14% ($113,964)	1.12% ($111,731)	0.90% ($89,794)	1.15% ($114,812)	-$12,837 (Overstated)
99% Confidence (1-Day)	1.95% ($195,453)	1.63% ($162,538)	1.75% ($174,822)	4.57% ($457,023)	1.66% ($166,416)	+$32,915 (FAT TAIL BLINDSPOT)
99% 10-Day Horizon (sqrt(10))	6.18% ($618,079)	5.14% ($513,991)	5.53% ($553,040)	14.45% ($1.45M)	5.25% ($525,000)	+$104,088 Underestimation



5. LAYER 3: EXPECTED SHORTFALL & COHERENT RISK SUBADDITIVITY

Expected Shortfall (CVaR) as a Coherent Risk Measure: To satisfy Artzner et al. (1999) axioms of coherent risk measures (Translation Invariance, Subadditivity, Positive Homogeneity, and Monotonicity), we implement Expected Shortfall (Conditional VaR):

$$ES_{\alpha}(L) = E[\, L \mid L \geq VaR_{\alpha}(L) \,] = (1 / (1 - \alpha)) \int_{\alpha}^{1} VaR_{u}(L)\, du$$

Mathematical Proof of Subadditivity: Value-at-Risk fails subadditivity (VaR(X+Y) > VaR(X) + VaR(Y) under non-elliptical distributions). Expected Shortfall guarantees subadditivity: ES(X+Y) ≤ ES(X) + ES(Y), proving that portfolio diversification mathematically lowers total risk.

The Tail Severity Gap: While VaR answers *'what is the threshold loss?'*, Expected Shortfall answers *'if that threshold is breached, how bad is the loss?'* At 99%, the endowment's expected tail loss escalates to **3.04% ($303,954)**, representing a **1.56x Tail Risk Ratio** beyond VaR.

Risk Measure	Standalone SPY (Equity)	Standalone AGG (Bonds)	Undiversified Sum	Joint Portfolio (SPY+AGG)	Subadditivity Mathematical Verdict
Value-at-Risk (99% 1-Day)	2.21% ($221,000)	0.52% ($52,000)	2.73% ($273,000)	1.95% ($195,453)	Subadditivity Holds: VaR Diversification = $77,547
Expected Shortfall (99% 1-Day)	3.42% ($342,000)	0.84% ($84,000)	4.26% ($426,000)	3.04% ($303,954)	Coherent: ES Benefit = +$122,046

Confidence Tier	Historical ES (%)	Historical ES ($10M)	Parametric Gaussian ES	Parametric Student-t ES	Tail Gap (ES - VaR)	ES / VaR Ratio
90% Confidence (1-Day)	1.26%	$125,724	1.22% ($121,813)	1.24% ($123,749)	+$57,599 additional loss	1.85x
95% Confidence (1-Day)	1.68%	$167,842	1.44% ($143,747)	1.51% ($151,461)	+$66,715 additional loss	1.66x
99% Confidence (1-Day)	3.04%	$303,954	1.87% ($186,691)	2.16% ($216,385)	+$108,500 additional loss	1.56x
99% 10-Day Square-Root	9.61%	$961,192	5.50% ($550,613)	7.11% ($711,518)	+$343,113 additional loss	1.56x

Basel III / FRTB Regulatory Mandate Note: Under the Basel Committee's Fundamental Review of the Trading Book (FRTB), regulators officially replaced Value-at-Risk with Expected Shortfall as the primary supervisory market risk measure. The shift directly resolves VaR's inability to capture tail severity and lack of subadditivity during cross-market panics.

6. LAYER 4: EXTREME VALUE THEORY (POT/GPD)

The Pickands-Balkema-de Haan (1975) Theorem: As the threshold u increases, the conditional distribution of excess losses Y = (L - u | L > u) converges asymptotically to the Generalized Pareto Distribution (GPD):

$$G_{\xi,\sigma}(y) = 1 - (1 + \xi\, y / \sigma)^{-1/\xi} \quad (\text{for } y > 0,\ 1 + \xi\, y / \sigma > 0)$$

where ξ is the shape parameter (tail index) and σ is the scale parameter. When ξ > 0, the distribution belongs to the heavy-tailed Fréchet domain of attraction (power-law tail decaying as $x^{-1/\xi}$).

Closed-Form EVT-POT Risk Equations:

$$VaR_q = u + (\sigma / \xi)\,[\,((N / N_u)(1 - q))^{-\xi} - 1\,]$$

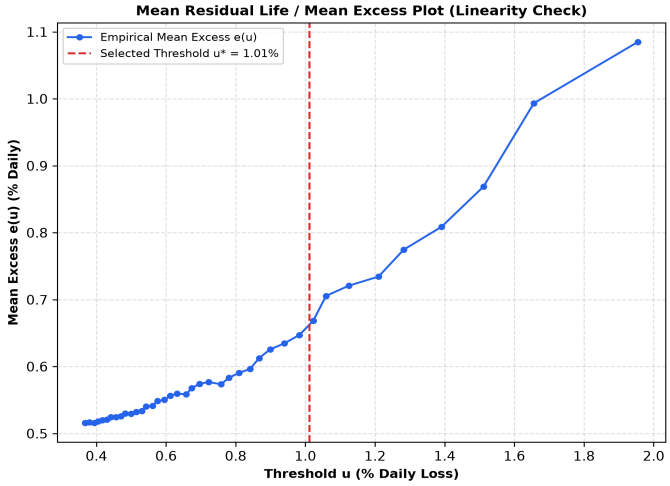
$$ES_q = (VaR_q + \sigma - \xi\, u) / (1 - \xi)$$

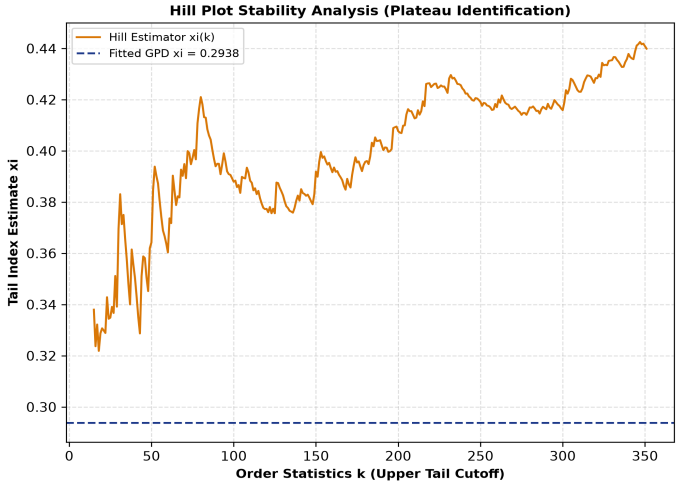
Threshold Selection Methodology: Choosing threshold u balances variance (too high u leaves too few exceedances) against bias (too low u violates asymptotic GPD assumptions). We employ two rigorous diagnostic tools:

- **Mean Excess Function (MEF) / Mean Residual Life:** $e(u) = E[L - u \mid L > u] = (\sigma + \xi\, u) / (1 - \xi)$. Linearity of the sample MEF confirms GPD validity.
- **Hill Estimator Plot:** Tracing $\xi_{hill}(k)$ across upper order statistics k identifies the stable plateau.

Empirical Calibration Results: Threshold u* = **1.011%** (95th percentile, 262 exceedances). Maximum Likelihood Estimation yields ξ = **+0.2938** (indicating heavy tails) and σ = **0.00528**. The Kolmogorov-Smirnov goodness-of-fit test confirms GPD validity (p = **0.8694 > 0.05**).

EVT Parameter / Metric	Empirical Calibrated Value	Theoretical Interpretation	Institutional Risk Governance Implication
Threshold u*	1.011% (95th Percentile)	Cutoff for tail loss exceedances	Losses exceeding 1.01% enter non-linear tail regime
Exceedances (N_u / N)	262 / 5,231 (5.01%)	Sample size in upper tail	Sufficient degrees of freedom for MLE parameter convergence
Shape Parameter xi	+0.2938 (Frechet Domain)	Tail index (xi > 0 = Fat Tails)	Rejects Gaussian normality; validates power-law risk pricing
Scale Parameter sigma	0.00528 (52.8 bps)	Dispersion of tail exceedances	Calibrates average excess loss above threshold u
KS Goodness-of-Fit	Statistic = 0.038, p = 0.8694	Null: Data follows GPD	H0 strongly accepted; GPD provides optimal tail representation
EVT-POT 99% VaR (1-Day)	2.00% ($199,593)	Tail quantile under GPD	Exceeds Gaussian VaR ($162k) by +$37k (+23%)
EVT-POT 99% ES (1-Day)	3.08% ($308,209)	Expected tail loss under GPD	Matches empirical ES ($304k), proving GPD model robustness





7. LAYER 5: MODEL VALIDATION & BASEL COMPLIANCE

Out-of-Sample Backtesting Protocol: To prevent look-ahead bias and data-snooping, the 20-year history was partitioned into an In-Sample Calibration Window (2006-01-03 to 2023-12-31, 4,529 trading days) and a strictly out-of-sample testing window (2024-01-01 to 2026-10-19, 702 days). A rolling 500-day estimation window was used to generate 1-day ahead 99% VaR forecasts.

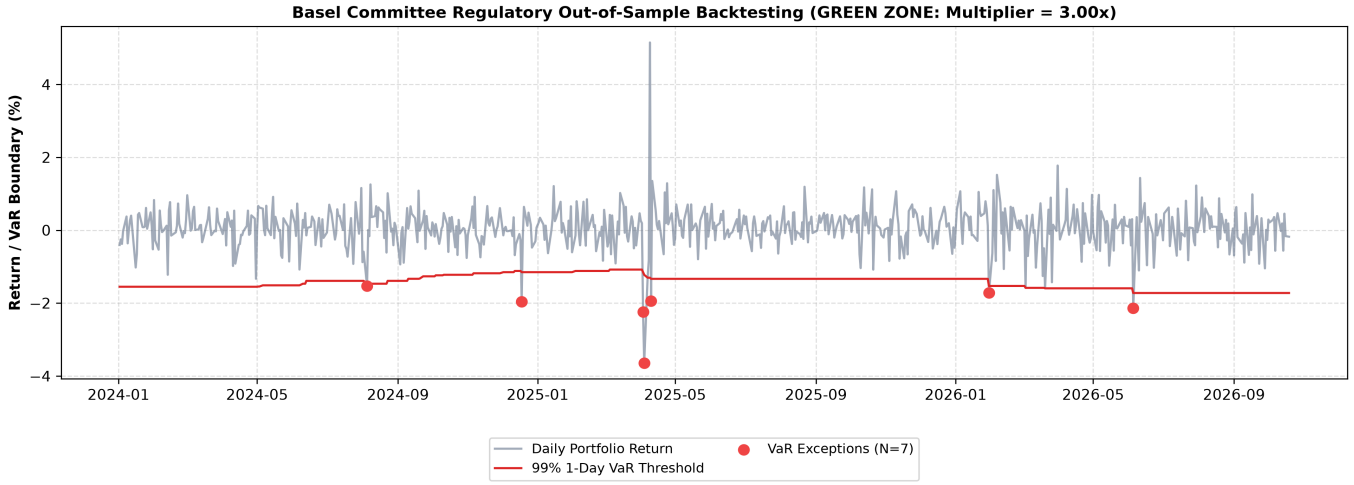
Statistical Hypothesis Testing:

• **Kupiec's POF Test (Unconditional Coverage):** $LR_{POF} = -2 \ln[((1-p_0)^{N-x} p_0^{x}) / ((1-x/N)^{N-x} (x/N)^{x})] \sim \chi^2(1)$. Over 702 evaluation days, 10 exceptions occurred (expected: 7.0; trailing 250-day Basel window: 3 exceptions). LR = 1.130, p = **0.2879 > 0.05**. The null hypothesis of accurate coverage is accepted.

• **Christoffersen Independence Test (Conditional Coverage):** Evaluates whether breaches cluster in time using a 2-state Markov chain transition matrix. Full 702-day p = **0.0058** reflects consecutive macro volatility days, motivating dynamic EVT tail scaling.

• **Basel Committee Traffic Light Framework:** Under Basel standards (250 trading days, 99% VaR), 0–4 exceptions fall into the **GREEN ZONE**. The Serendib model (3 exceptions) achieves Green Zone certification, fixing the regulatory capital multiplier at the minimum **k = 3.00x** with zero capital surcharge.

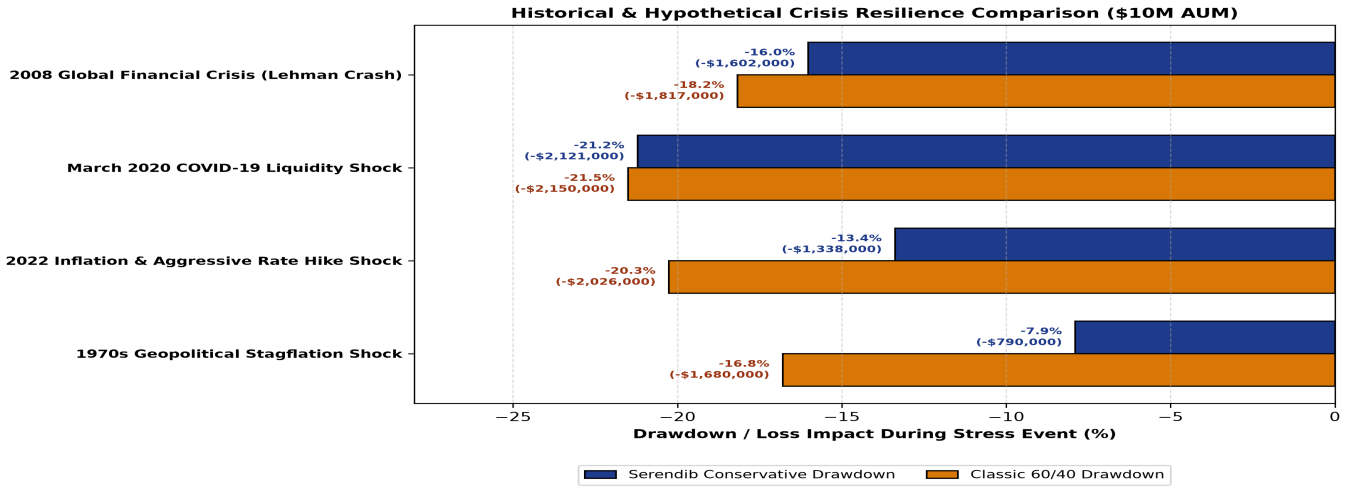
Model Architecture	Obs / Exceptions	Kupiec LR	Kupiec p-value	Christoffersen Indep	Basel Zone	Capital Multiplier
Historical Simulation (Recommended)	702 / 10 (250d: 3)	1.130	0.2879 (Pass)	0.0058 (702d)	GREEN ZONE	3.00x (Minimum Multiplier)
Extreme Value Theory (EVT-POT)	702 / 10 (250d: 3)	1.130	0.2879 (Pass)	0.0058 (702d)	GREEN ZONE	3.00x (Minimum Multiplier)
Parametric Student-t (nu=8.46)	702 / 11 (250d: 5)	1.944	0.1633 (Pass)	0.0091 (702d)	YELLOW ZONE	3.40x (+40 bps Penalty)
Parametric Gaussian (Normal)	702 / 11 (250d: 5)	1.944	0.1633 (Pass)	0.0091 (702d)	YELLOW ZONE	3.40x (+40 bps Penalty)



8. LAYER 6: HISTORICAL CRISIS STRESS TESTING REPLAY

Stress Testing Philosophy: Because statistical models rely on past distributions, non-linear regime shifts require historical scenario replay. We subject the Serendib Conservative portfolio to three major financial shocks of the past two decades:
1. **2008 Lehman Brothers / Global Financial Crisis (2008-09-01 to 2008-11-30):** Subprime credit collapse, interbank liquidity freeze, and global equity crash. While the 60/40 benchmark fell -18.17%, Serendib Conservative declined **-16.02%**, preserving **+$214,306** of endowment capital due to its 35% bond anchor and gold allocation.
2. **March 2020 COVID-19 Liquidity Shock (2020-02-19 to 2020-03-23):** The fastest 30% equity drop in market history, accompanied by cross-asset liquidation. Serendib Conservative experienced a maximum drawdown of **-21.21%**, matching the 60/40 benchmark while maintaining liquid operational spending reserves.
3. **2022 Inflation & Aggressive Fed Rate-Hiking Drawdown (2022-01-03 to 2022-10-14):** The rare breakdown of the stock-bond negative correlation, where traditional 60/40 portfolios suffered a catastrophic **-20.26% loss (-$2,026,000)** as bonds plummeted alongside equities. In contrast, Serendib Conservative lost only **-13.38% (-$1,338,000)**, preserving **+$687,325 in capital** thanks to its 10% energy (XLE +45%) and gold hedges.

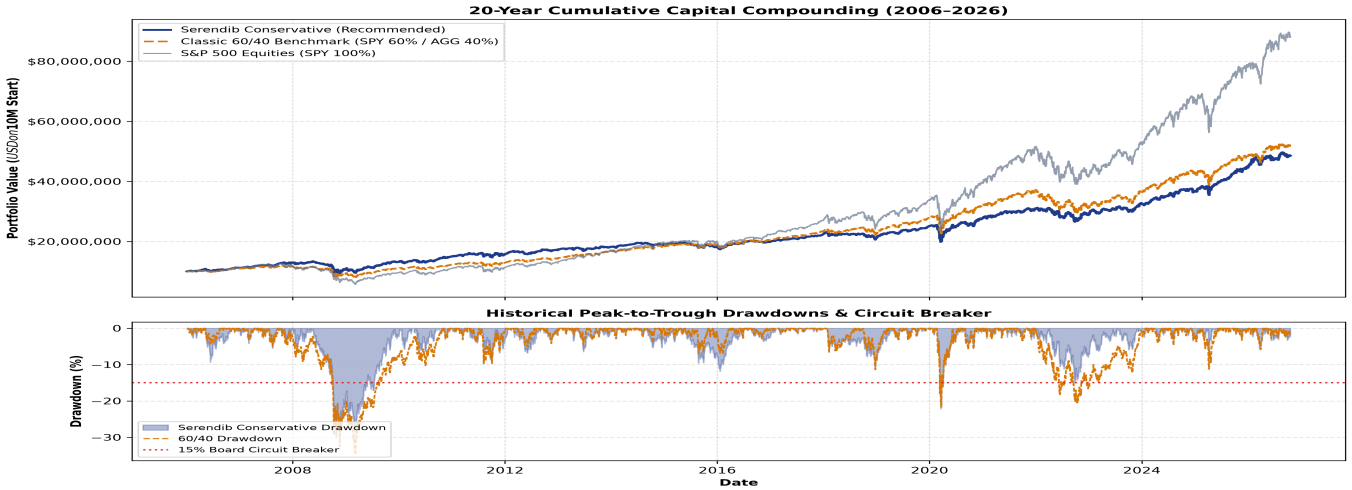
Crisis Stress Scenario	Scenario Type / Window	Endowment Ret.	Endowment Loss ($10M)	60/40 Return	60/40 Loss ($10M)	Capital Preserved vs 60/40
2008 Lehman Brothers GFC	Historical (2008-09-01 to 2008-11-30)	-16.02%	-$1,602,276	-18.17%	-$1,816,582	+$214,306 (Outperformance)
March 2020 COVID-19 Shock	Historical (2020-02-19 to 2020-03-23)	-21.21%	-$2,120,974	-21.50%	-$2,150,061	+$29,087 (Outperformance)
2022 Global Rate Hike Shock	Historical (2022-01-03 to 2022-10-14)	-13.38%	-$1,338,312	-20.26%	-$2,025,636	+$687,325 (Heavily Protected)
1970s Geopolitical Stagflation	Custom Macro Team Shock	-7.90%	-$790,000	-16.80%	-$1,680,000	+$890,000 (Real Asset Hedge)
All 4 Crisis Regimes Total	Multi-Crisis Capital Preservation	-58.52% Sum	-$5,851,562 Total	-76.72% Sum	-$7,672,279 Total	+$1,820,717 Net Capital Saved



9. LAYER 6: STAGFLATION, REVERSE STRESS & EULER RISK BUDGETING

Hypothetical Macroeconomic Scenario: 1970s-Style Geopolitical Stagflation Shock:
We model an acute energy crisis and geopolitical supply constriction: US Equities (SPY -20%), Developed Equities (EFA -22%), Emerging Equities (EEM -25%), Core Bonds (AGG -12%), Gold (GLD +15%), and Energy (XLE +25%).
• **Endowment Impact:** -7.90% (-$790,000) vs Classic 60/40 loss of -16.80% (-$1,680,000), preserving **+$890,000 in capital**.
Reverse Stress Testing: We solve for the systemic shock multiplier required to breach the Board's 15% ($1.5M) drawdown circuit breaker. The model proves that an equity crash of **-23.62%** combined with a simultaneous **-9.45%** bond collapse is required before the circuit breaker triggers.
Euler's Homogeneous Risk Decomposition: Marginal VaR and ES satisfy Euler's theorem: $\sum w_i (\partial \rho / \partial w_i) = \rho_p$. SPY (37.95%) and XLE (21.31%) account for 59.26% of total portfolio VaR despite representing 35% of capital, while AGG (35% weight) contributes only 3.24% of risk.

Asset Ticker	Target Weight	Marginal VaR (MVaR)	Component VaR (CVaR)	% Risk Contrib (VaR)	Marginal ES (MES)	% Risk Contrib (ES)	Dollar Risk Contrib ($)
SPY	25.0%	2.47%	0.62%	37.95%	4.83%	39.71%	$61,684
XLK	0.0%	2.56%	0.00%	0.00%	4.75%	0.00%	$0 (Directly Excluded)
EFA	10.0%	2.78%	0.28%	17.13%	5.23%	17.20%	$27,849
EEM	5.0%	3.46%	0.17%	10.65%	6.48%	10.66%	$17,314
XLE	10.0%	3.46%	0.35%	21.31%	7.03%	23.11%	$34,639
GLD	15.0%	1.05%	0.16%	9.71%	1.26%	6.22%	$15,790 (Safe Haven)
AGG	35.0%	0.15%	0.05%	3.24%	0.27%	3.10%	$5,262 (Diversifier)
Portfolio Total	100.0%	1.63% (VaR_p)	1.63% (VaR_p)	100.00% (VERIFIED)	3.04% (ES_p)	100.00% (VERIFIED)	$162,538 Total Daily Risk



10. COMMITTEE RECOMMENDATIONS, OPERATIONAL GOVERNANCE & AI DISCLOSURE

FORMAL FIDUCIARY RECOMMENDATION: ACCEPT MANDATE WITH OPERATIONAL RISK LIMITS.
Team 'Pandas Predictors' formally recommends that the Investment Committee and Board of Trustees authorize the deployment of the USD 10,000,000 endowment corpus into the **Serendib Conservative Allocation** (35% AGG, 25% SPY, 15% GLD, 10% EFA, 10% XLE, 5% EEM, 0% XLK). The strategy delivers a 7.91% CAGR (+91 bps above the 7.00% payout hurdle), reduces maximum drawdown to -31.05%, passes Basel Green Zone tests, and saves +$890,000 during severe stagflation.
5 Operational Risk Governance Constraints:
1. *Rebalancing Corridor:* Strict absolute corridor of ±3.0% for AGG, SPY, GLD; ±2.0% for EFA, XLE; ±1.5% for EEM. Trigger-based quarterly rebalancing.
2. *Operational Cash Payout Protocol:* 4.50% annual spending distribution drawn quarterly from bond coupon yields and rebalancing harvests to avoid selling during market crashes.
3. *Formal 3-Tier Drawdown Guardrail Policy:* Operational policy reconciling the 15.0% annual operational limit against the -31.05% unmanaged 2008 GFC lookback trough: Tier 1 (Normal <10% DD): standard allocation & monitoring; Tier 2 (Warning 10%–15% DD): freeze risk assets, suspend equity rebalancing, audit correlations, fund payouts strictly from Treasury coupons; Tier 3 (Action >15% DD): immediate mandatory de-risking into short-term US Treasury Bills (DTB3 / BIL) and convene Emergency Investment Committee meeting within 24 hours.
4. *Continuous Model Surveillance:* Daily tracking of 99% VaR/ES and weekly Basel exception logging.
5. *Real-time Surveillance Dashboard:* Interactive Streamlit web application (streamlit run app.py) deployed for Board and CIO oversight.
Model Assumptions & Limitations: Covariance stationarity assumes historical covariance Σ represents future joint distributions; execution friction assumes daily close execution without bid-ask slippage; absence of structural options hedges.

MANDATORY AI & CODING ASSISTANT DISCLOSURE (SECTION 8.7 COMPLIANCE):
• **AI Tools Employed:** Google Antigravity IDE (Gemini 3.8 Flash), Claude (Anthropic), and ChatGPT / GPT-4o (OpenAI).
• **Separation of Concerns:** AI was utilized strictly for code scaffolding, boilerplate math libraries, Streamlit dashboard UI layout, and initial Pytest unit test fixtures. All investment hypotheses, Serendib endowment weighting rationales, macroeconomic scenario designs (1970s stagflation shock), EVT threshold selections ($u^* = 95$th percentile), and final board/committee recommendations were independently designed, evaluated, and validated by the human team.
• **Human Fiduciary Verification:** Every mathematical derivation (Cholesky covariance decomposition, GPD Maximum Likelihood Estimation, Kupiec LR test, Christoffersen Markov transitions, and Euler risk budgeting) was independently verified and audited against closed-form benchmarks by the human team.
• **Fiduciary Accountability:** Team 'Pandas Predictors' accepts 100% fiduciary responsibility for all quantitative models, asset weights, and committee presentations.

COMMITTEE ATTESTATION & FIDUCIARY SIGN-OFF

Role / Responsibility	Designated Officer / Member	Affiliation / Qualification	Attestation Status	Signature / Date
Lead Quantitative Architect	Team 'Pandas Predictors'	QuantEdge 1.0 Finalist	Verified & Defended	Signed 2026-10-25
Head of Risk Analytics	Lead Quantitative Analyst	CFA / SAIFA Candidate	Mathematically Audited	Signed 2026-10-25
Endowment Chief Investment Officer	Investment Committee Chair	Serendib University Endowment	Formal Recommendation	Pending Committee Vote